

INFOSYS SPRINGBOARD VIRTUAL INTERNSHIP 7.0

Architecture On

AI-Assisted Threat Detection Dashboard Using AWS Cloud and Machine Learning

Submitted in Partial Fulfillment of the Requirements

for the

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Submitted By:

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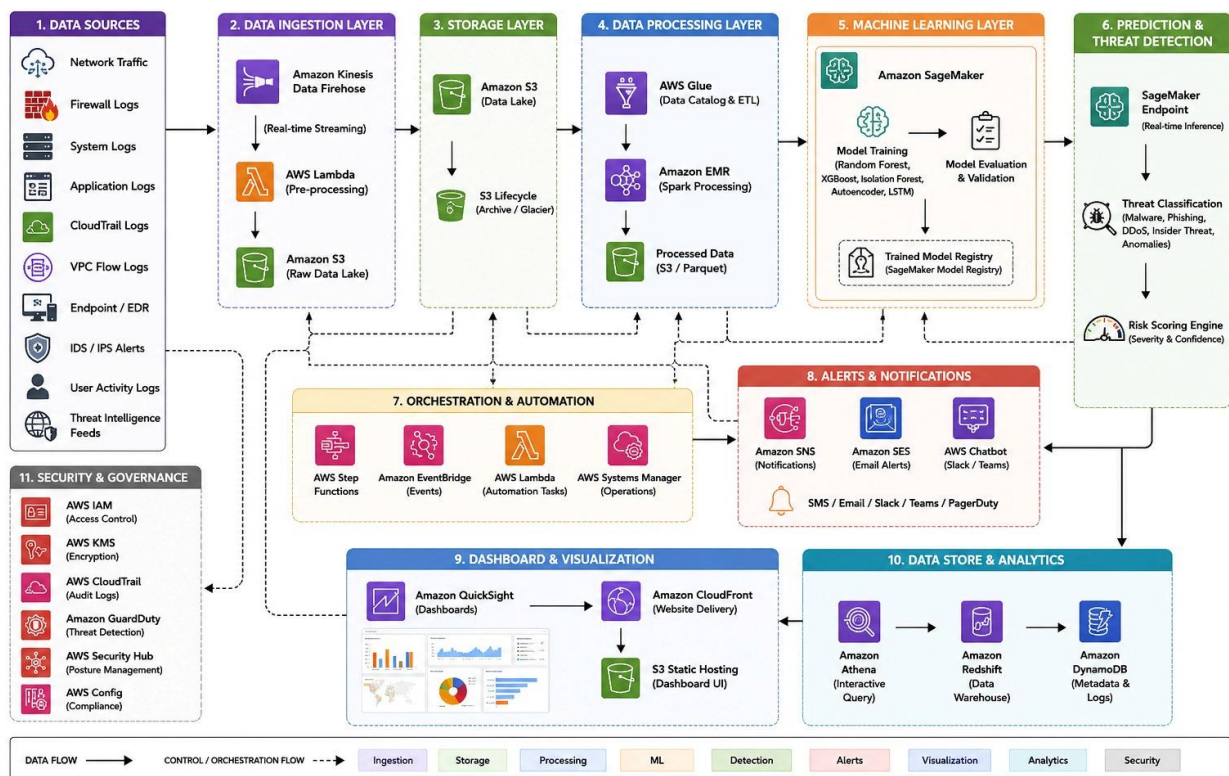
ARCHITECTURE DESCRIPTION

The proposed AI-Assisted Threat Detection Dashboard Using AWS Cloud and Machine Learning provides a scalable and intelligent solution for detecting cyber threats in real time. It collects security data from sources such as network traffic, firewall logs, CloudTrail, VPC Flow Logs, and endpoint devices. The data is ingested using Amazon Kinesis and AWS Lambda, then securely stored in Amazon S3 for further processing.

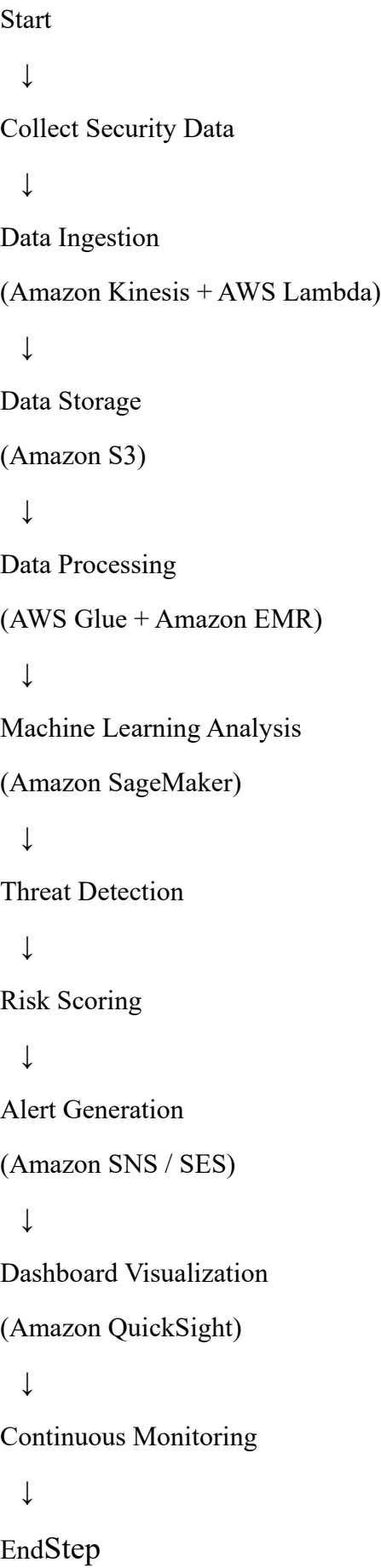
The stored data is cleaned and transformed using AWS Glue and Amazon EMR, after which Amazon SageMaker applies machine learning algorithms such as Random Forest, XGBoost, Isolation Forest, and LSTM to detect anomalies and classify cyber threats. The system generates risk scores and predicts the severity of each detected threat.

Automated workflows managed by AWS Step Functions, EventBridge, and Lambda trigger alerts through Amazon SNS, SES, and AWS Chatbot. Security insights, threat trends, and reports are displayed on an interactive Amazon QuickSight dashboard, while IAM, KMS, CloudTrail, GuardDuty, and Security Hub ensure secure access, monitoring, and compliance. Overall, the architecture enables accurate, real-time threat detection, automated response, and effective cybersecurity management.

AI-Assisted Threat Detection Dashboard Using AWS Cloud and Machine Learning



WORKFLOW STEPS



1. COLLECT SECURITY DATA

- Gather logs from network traffic, firewall, CloudTrail, VPC Flow Logs, system logs, application logs, and endpoint devices.

2. INGEST DATA

- Stream the collected data using Amazon Kinesis Data Firehose and preprocess it with AWS Lambda.

3. STORE DATA

- Save raw and processed security logs in Amazon S3, with lifecycle policies for long-term storage.

4. PROCESS DATA

- Clean, transform, and normalize data using AWS Glue and Amazon EMR to prepare it for analysis.

5. TRAIN AND DEPLOY ML MODELS

- Use Amazon SageMaker to train machine learning models such as Random Forest, XGBoost, Isolation Forest, and LSTM for threat detection.

6. Detect Threats

- Perform real-time inference using SageMaker Endpoints to identify anomalies, malware, phishing, DDoS attacks, and insider threats.

7. GENERATE RISK SCORES

- Classify detected threats and assign severity and confidence scores based on their potential impact.

8. AUTOMATE RESPONSE

- Coordinate workflows using AWS Step Functions, EventBridge, and Lambda to trigger predefined response actions.

9. SEND ALERTS

- Notify security teams through Amazon SNS, Amazon SES, and AWS Chatbot via email, SMS, Slack, or Microsoft Teams.

10. VISUALIZE RESULTS

- Display real-time threats, analytics, and reports using Amazon QuickSight dashboards.

11. ENSURE SECURITY AND GOVERNANCE

- Use AWS IAM, AWS KMS, CloudTrail, GuardDuty, Security Hub, and AWS Config to provide secure access, monitoring, and compliance.

12. CONTINUOUS MONITORING AND IMPROVEMENT

- Store historical data, retrain machine learning models with new threat patterns, and continuously improve detection accuracy.

TECHNOLOGIES USED

Category	Technology
Cloud Platform	AWS (Amazon Web Services)
Programming Language	Python
Frontend	HTML, CSS, JavaScript
Backend	Flask / FastAPI
Data Storage	Amazon S3
Data Processing	AWS Glue, Amazon EMR
Machine Learning	Amazon SageMaker
ML Algorithms	Random Forest, XGBoost, Isolation Forest, LSTM
Dashboard	Amazon QuickSight
Alerts	Amazon SNS, Amazon SES
Security	IAM, KMS, CloudTrail, GuardDuty
Version Control	Git, GitHub